

Project Design Phase
Proposed Solution Template

Date	2 nd jul 2026
Team ID	
Project Name	NUTRTION ASSISTANT
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Many individuals struggle to maintain a balanced diet due to the lack of personalized nutrition guidance, inefficient meal planning, and difficulty tracking calorie and nutrient intake. Dietitians also face challenges in managing multiple clients and monitoring their progress efficiently using manual methods.
2	Idea / Solution Description	Develop a Nutrition Assistant Application using the MERN Stack (MongoDB, Express.js, React.js, Node.js) that enables users to register securely, create personalized profiles, receive customized meal plans, track daily calories and nutrients, monitor health progress, and access nutritional insights. Dietitians can manage clients, prepare meal plans, and monitor progress through an interactive dashboard, while administrators oversee users and system operations with role-based access control.
3	Novelty / Uniqueness	The application combines personalized meal planning, nutrient analysis, calorie tracking, client management, and progress monitoring into a single secure platform. Role-based dashboards for users, dietitians, and administrators provide a comprehensive digital nutrition management system with real-time tracking and data visualization.
4	Social Impact / Customer Satisfaction	The solution encourages healthier eating habits, improves awareness of nutrition, supports disease prevention through better diet management, and helps users achieve their fitness goals. It also improves the efficiency of dietitians by reducing manual work and providing better client engagement, leading to higher customer satisfaction.
5	Business Model (Revenue Model)	The application can adopt a freemium subscription model, where basic features such as meal tracking and calorie monitoring are free, while premium features like personalized dietitian consultations, advanced nutrition reports, AI-based recommendations, and detailed analytics are available through monthly or yearly subscriptions. Additional revenue can be generated through partnerships with fitness centers, healthcare providers, and healthy food brands.
6	Scalability of the Solution	The system is designed using a modular MERN architecture, making it easy to scale by adding new features such as AI-powered meal recommendations, wearable device integration, multilingual support,

		mobile applications, teleconsultation with dietitians, and cloud deployment. The architecture supports a growing number of users while maintaining performance and security.
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